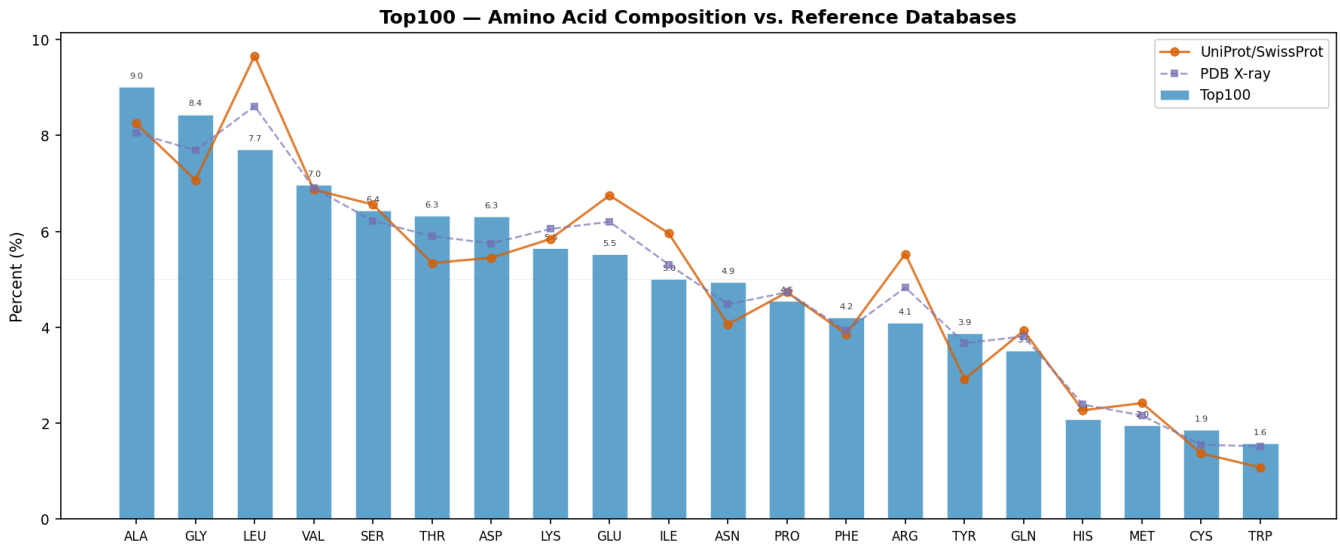
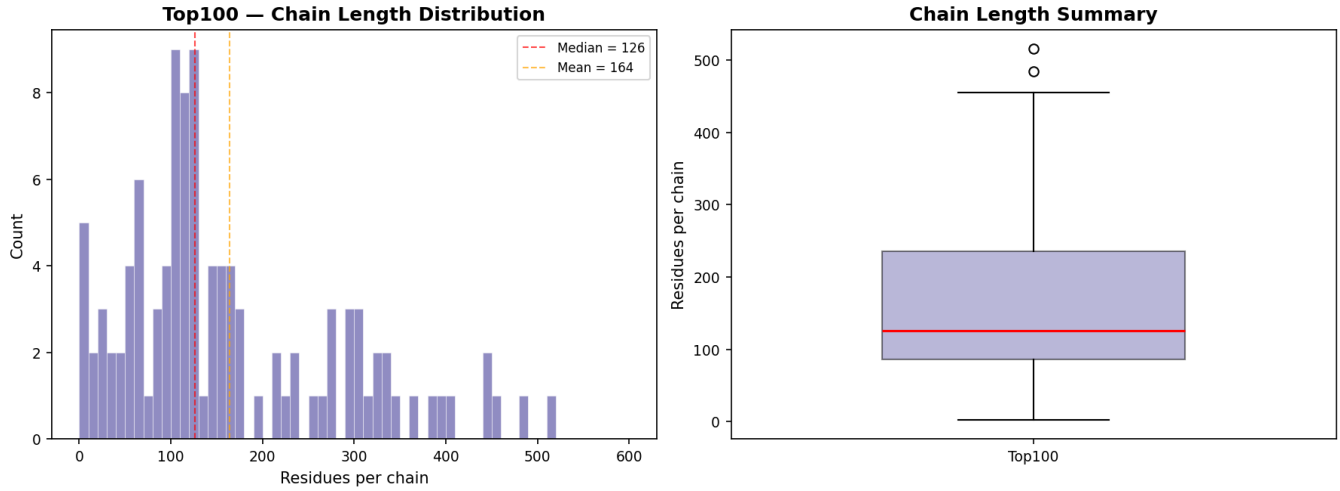
Top100 Dataset — Geometric Analysis

Residues: 17,434 | **Source:** top100_measures.jsonl | **Generated by:** pydangle-biopython v0.5.1

```
phi: 17,218  psi: 17,218  omega: 17,218  tau: 17,434
chi1: 14,380  chi2: 10,616  chi3: 6,441  chi4: 3,099
```



Residue	Count	Dataset%	UniProt%	PDB%	Δ UniProt
ALA	1,570	9.01%	8.25%	8.05%	+0.76%
GLY	1,471	8.44%	7.07%	7.69%	+1.37%
LEU	1,343	7.70%	9.66%	8.61%	-1.96%
VAL	1,215	6.97%	6.87%	6.90%	+0.10%
SER	1,122	6.44%	6.56%	6.22%	-0.12%
THR	1,102	6.32%	5.34%	5.90%	+0.98%
ASP	1,099	6.30%	5.45%	5.75%	+0.85%
LYS	985	5.65%	5.84%	6.05%	-0.19%
GLU	963	5.52%	6.75%	6.20%	-1.23%
ILE	872	5.00%	5.96%	5.31%	-0.96%
ASN	862	4.94%	4.06%	4.48%	+0.88%
PRO	792	4.54%	4.73%	4.73%	-0.19%
PHE	732	4.20%	3.86%	3.93%	+0.34%
ARG	714	4.10%	5.53%	4.83%	-1.43%
TYR	676	3.88%	2.92%	3.67%	+0.96%
GLN	613	3.52%	3.93%	3.81%	-0.41%
HIS	364	2.09%	2.27%	2.39%	-0.18%
MET	340	1.95%	2.42%	2.16%	-0.47%
CYS	325	1.86%	1.37%	1.55%	+0.49%
TRP	274	1.57%	1.08%	1.52%	+0.49%

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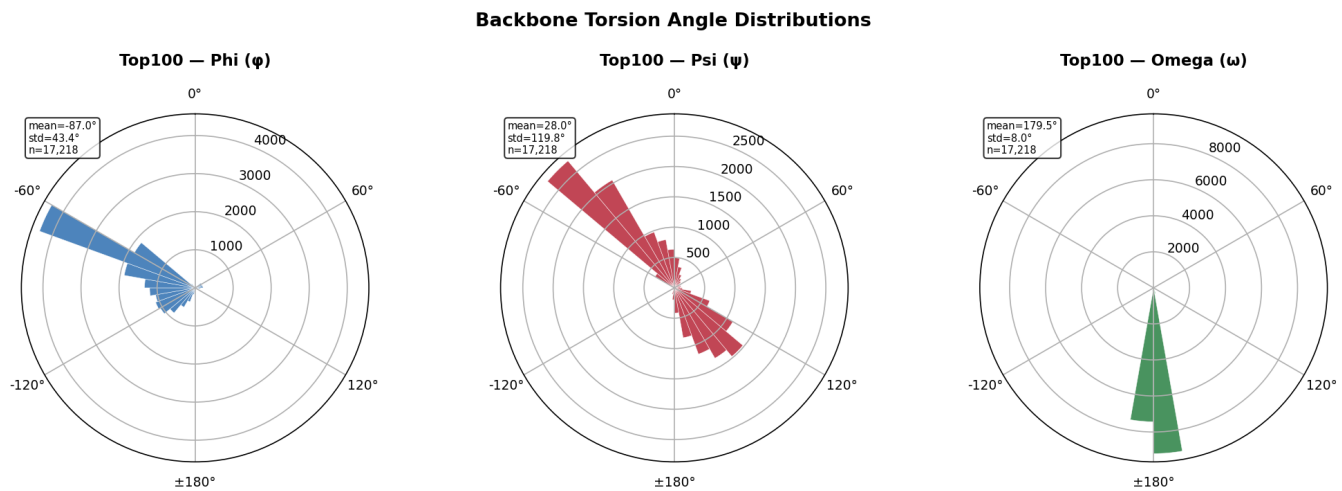
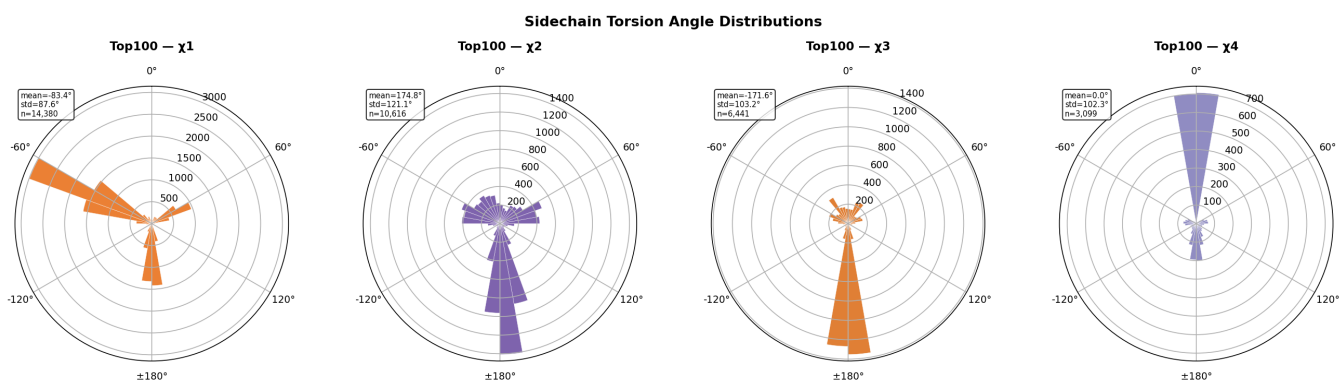
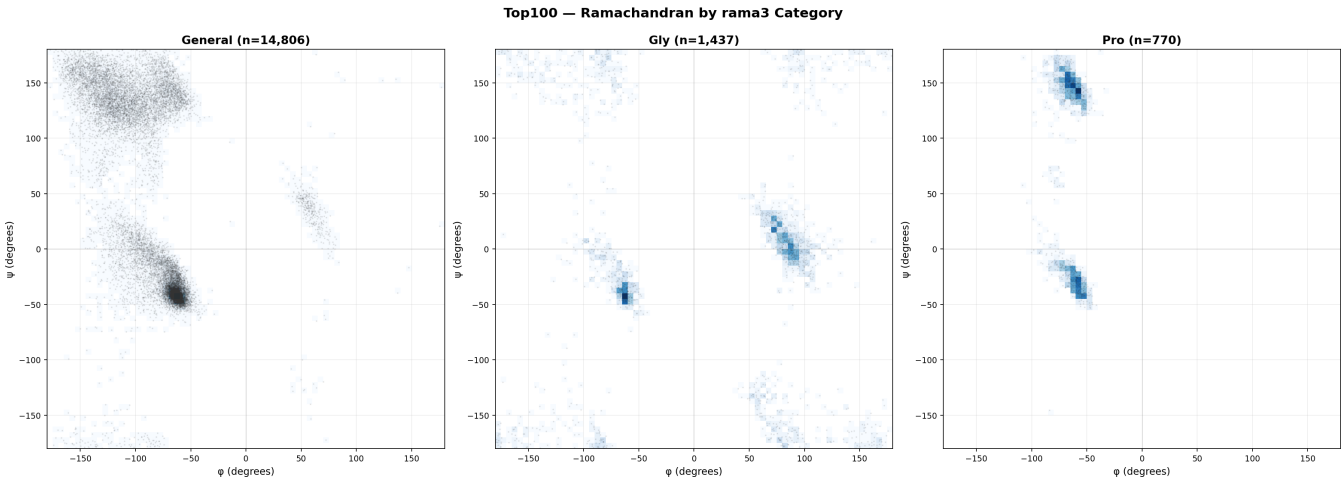
Chain length statistics (106 chains):
  Min:              2
  Q1:              86
  Median:         126
  Mean:         164.5
  Q3:             235
  Max:            516
  Std:          119.8
```





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Category counts (rama3):
  General        14,806  (84.93%)
  Gly             1,437  ( 8.24%)
  Pro               770  ( 4.42%)
```

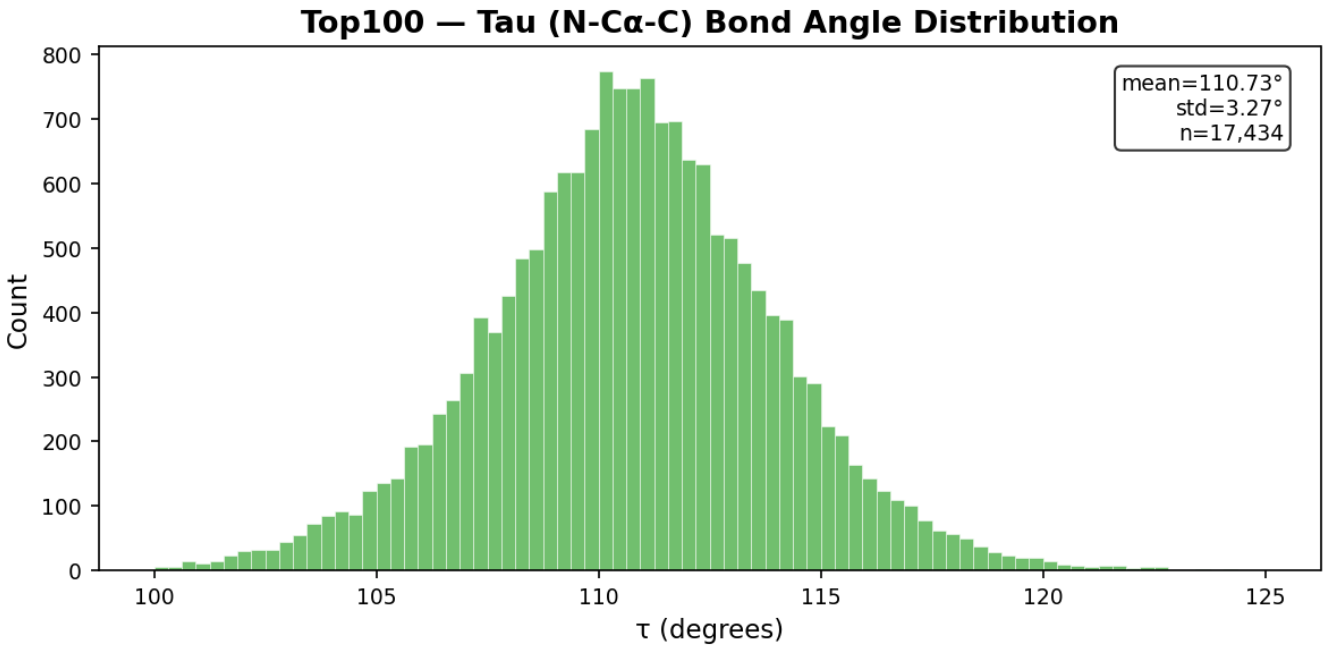

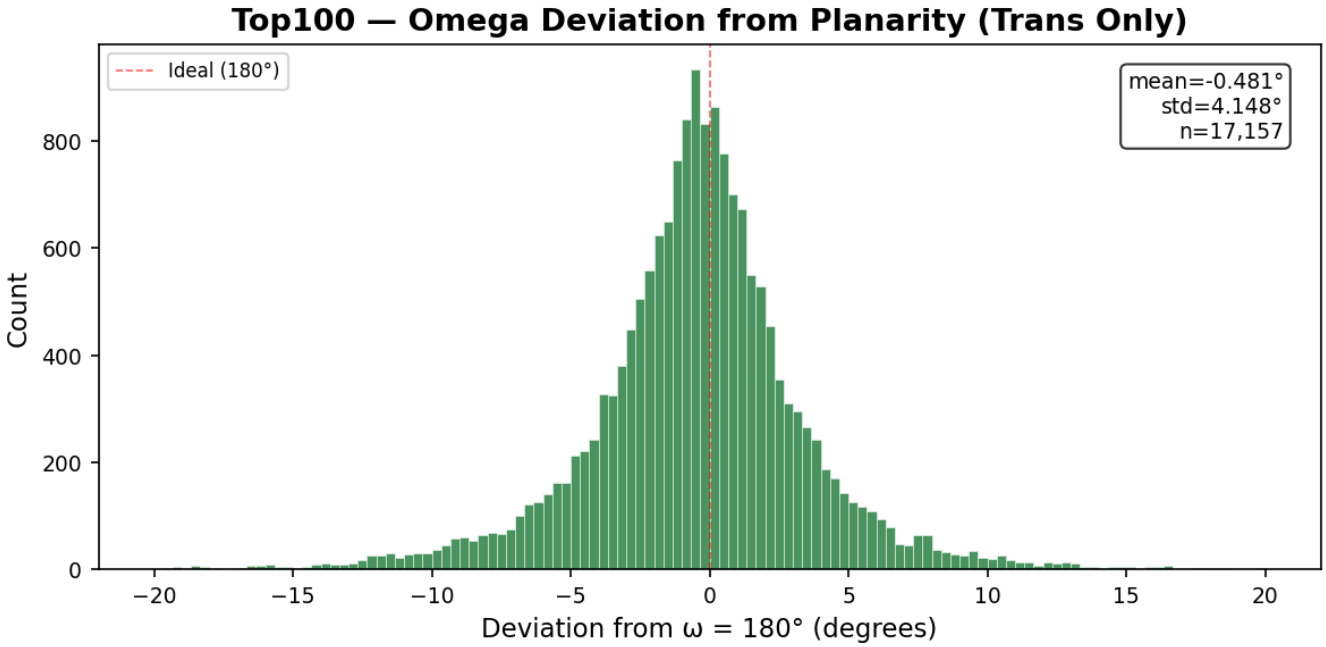


```
Dataset:                   Top100
Total residues:            17,434
Unique structures:         98
Unique chains:             106

Ramachandran category distribution:
  General        12,187  (69.90%)
  IleVal          2,052  (11.77%)
  Gly             1,437  ( 8.24%)
  TransPro          720  ( 4.13%)
  PrePro            567  ( 3.25%)
  CisPro             50  ( 0.29%)

Angle             N    Circ Mean     Circ Std
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phi          17,218      -86.970       43.398
psi          17,218       27.990      119.824
omega        17,218      179.521        7.966
tau          17,434      110.729        3.269
chi1         14,380      -83.355       87.557
chi2         10,616      174.761      121.133
chi3          6,441     -171.584      103.209
chi4          3,099        0.024      102.314
```